Project Proposal

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EE-222 Microprocessor Systems

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Project Statement

Implementation of a complete hardware/software integrated microcontroller based embedded system. The system should be implemented in assembly/C/C++ language using Microchip MPLAB-X IDE and meet following specifications:

- **Microcontroller** (PIC18, ATMEGA32/ATMEGA16)
- At-least **two output devices** are mandatory, one should be LCD/7 segment display/OLED display, second can be buzzer, relays, motors etc.
- At-least **two input devices** are mandatory (Keypad/sensors), one should be connected with ADC, second can be switches etc
- **Battery operated** (mandatory)

Project Title

AVR-Based Solar Battery Monitoring and Control System with WIFI Connectivity

Description

This project aims to develop an embedded system for real-time monitoring and control of a solar-powered battery system using the **ATmega32** microcontroller with **WiFi connectivity**. The system monitors battery voltage, charge current, temperature, and light intensity to protect the battery from damage.

The system features remote monitoring through an **ESP8266 WiFi module**, allowing users to view real-time data on a web interface. An **LDR** measures ambient light, **LM35** monitors battery temperature, and **ACS712** measures current flow. All parameters are displayed on an **OLED display**, and control actions are executed through relay modules to prevent overcharging and deep discharge.

Components

Microcontroller:

- ATmega32 (AVR, 10-bit ADC, timers, UART, SPI, etc.)

Sensors & Modules:

- Voltage Divider (4kΩ and 1kΩ for 0-15V measurement)
- ACS712 Current Sensor (for charge/discharge current)
- LM35 Temperature Sensor (for battery thermal monitoring)
- LDR (Light Dependent Resistor)
- ESP8266 WiFi Module (for IoT connectivity)

Display & Interface:

- 128x64 OLED Display (SSD1306, I2C)
- Push buttons / Rotary encoder (for UI control)

Output Devices:

- 2-Channel Relay Module (for solar and load control)
- Buzzer (for audible alerts)

Power Supply:

- Solar Panel (10W-50W, 12V-18V)
- 12V Lead-Acid Battery
- DC-DC Buck Converter (12V to 5V)

Miscellaneous:

- Breadboard / PCB
- Jumper wires, resistors, capacitors
- Multimeter
- DC Load (fan/light bulb for testing)

Functionality Overview

1. **Sensing:** ATmega32 reads battery voltage (via divider), current (ACS712), temperature (LM35), and light intensity (LDR) through ADC channels.
2. **Control:** Implements charge control with overvoltage protection (cut-off at 14.4V) and discharge protection (load disconnect at 11.5V).
3. **Local Display:** OLED shows voltage, current, temperature, light level, and system status.
4. **Remote Monitoring:** ESP8266 sends data to cloud for web/mobile viewing.